

Polymer-Based Marine Antifouling and Fouling Release Surfaces: Strategies for Synthesis and Modification

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Abstract

In marine industries, the accumulation of organic matter and marine organisms on ship hulls and instruments limits performance, requiring frequent maintenance and increasing fuel costs. Current coatings technology to combat this biofouling relies heavily on the use of toxic, biocide-containing paints. These pose a serious threat to marine ecosystems, affecting both target and nontarget organisms. Innovation in the design of polymers offers an excellent platform for the development of alternatives, but the creation of a broad-spectrum, nontoxic material still poses quite a hurdle for researchers. Surface chemistry, physical properties, durability, and attachment scheme have been shown to play a vital role in the construction of a successful coating. This review explores why these characteristics are important and how recent research accounts for them in the design and synthesis of new environmentally benign antifouling and fouling release materials.

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INTRODUCTION TO BIOFOULING

Biofouling:

accumulation and growth of organic matter and aquatic organisms to a surface, usually to the detriment of the surface's function

Biofilm: a community of single and multicellular microorganisms in a matrix of extracellular polymeric substances adhered to a surface; grows on both living and nonliving surfaces

Diatom:

a single-celled alga that possesses a silica-based cell wall, usually found in biofilms

Cyprid: final larval stage before a barnacle reaches adulthood, responsible for exploration and identification of surfaces for settlement

Biocide: a toxic or poisonous substance used for the control of harmful or damaging organisms

At its most basic, biofouling is the accumulation and growth of biomolecules and organisms on surfaces, usually to the detriment of that surface's function. In marine environments it is a wide-reaching problem, impacting everything from ship maintenance to ecosystem safety. When a surface first enters a marine environment, the start of the fouling process is immediate, with the adsorption of dissolved organic matter, like proteins, carbohydrates, and proteoglycans, occurring in under a minute (1). This film conditions the surface to make it more hospitable to fouling organisms. The subsequent fouling species are often categorized into two main categories: soft and hard foulers (2). Soft foulers can be further divided into microfoulers, which form biofilms, generally composed of bacteria and diatoms, and macrofoulers like algae. Hard foulers, such as barnacles and mussels, are distinguished by their use of adhesives that harden to a tough cement-like material. Biofouling is often thought of as a successional process, with settlement and growth of foulers occurring in four stages: conditioning with proteins and carbohydrates, soft microfouler settlement and biofilm formation, soft macrofouler settlement, and then hard fouler settlement. Despite this, many foulers possess the ability to adhere to pristine surfaces, like motile zoospores of the algae *Ulva linza* and cyprids of several species of barnacles, which are often used as test organisms (3, 4).

The growth of these organisms on marine surfaces has both financial and environmental effects. Biofouling roughens ship hulls, leading to increased drag, limiting top speeds, and requiring higher fuel consumption, while also contributing to issues with instrument function and ship maneuverability (5). A 2011 study found that typical levels of biofouling cost the US Navy an estimated \$56 million a year in fuel penalties and maintenance for the entire DDG-51 ship class, with heavier fouling increasing those costs to \$119 million (6). Beyond the added costs to ship function, fouled ships can serve as transfer mechanisms for invasive species, putting local ecosystems at risk (7–10). In the past few years, Singapore has seen explosive growth of the invasive marine mussel *Mytella strigata*. First identified in March of 2016, by March of 2017 juveniles were found in extremely high densities, with reports of more than 10,000 per 100 cm² in some areas (11). With this came a radical decrease in documentation of the native green mussel *Perna viridis*. Identification of coatings that can help to alleviate these issues is highly desirable.

Historically, a variety of methods were used to combat biofouling, spanning everything from tar paints to copper panel siding (12). In the twentieth century, biocidal-based paints were the most successful, slowly leaching out a biocide to kill anything that settled. The most successful of these biocides were based on alkyl tins, most notably tributyl tin. Although incredibly effective, long-term use revealed a host of unintended environmental impacts. Tributyl tin was recognized as an endocrine disruptor to mollusks, causing the growth of male genitalia in female mussels, in the 1980s (13). It was also linked to abnormal shell growth in oysters and resulted in an estimated loss of more than \$147 million to the French oyster industry because of reduced oyster production in the late 1970s and early 1980s (14). Tributyl tin ultimately fell out of favor owing to increased regulation, and it was banned in 2008 by a convention set by the International Maritime Organization (15–17). It has since been replaced with other metal biocides, predominantly copper- and zinc-based systems, which, although effective, are facing increased scrutiny themselves because of buildup in harbors and marinas (18–20).

Although there is still ongoing research on alternative nonmetallic biocides, there is a very strong interest in developing entirely nontoxic coatings, as there is little control over how biocides affect nontarget organisms (21–24). For these coatings to succeed, they must work against a complex and varied marine environment. Different fouling organisms rely on different mechanisms of attachment, making the design of a universal coating incredibly difficult. And with

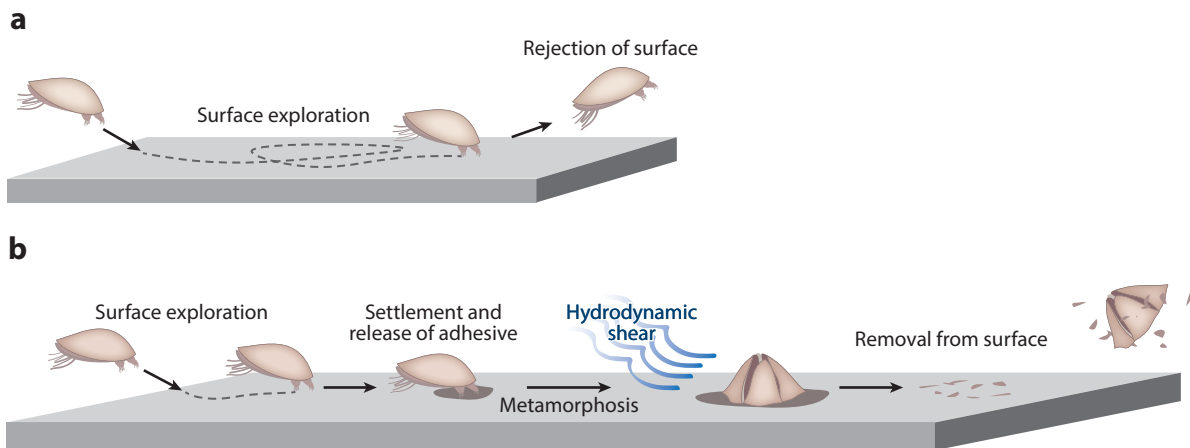


Figure 1

Schematic of mechanisms to combat marine biofouling. (a) Antifouling coatings present a surface that foulers do not find appealing for settlement. For example, a barnacle cyprid will land on the surface, and after exploring it will decide it is not hospitable and search elsewhere to settle. (b) Fouling release coatings present a surface that fouling organisms can settle on but cannot firmly attach to. Here a barnacle cyprid explores the surface and finds it acceptable to settle on, releasing an adhesive and metamorphosing into an adult barnacle. But its attachment is weak, so with hydrodynamic shear from water jet cleaning, or a ship's motion through the water, it is easily removed.

marine vessels often traveling cross-continentially, ensuring that a coating can perform against a wide array of organisms in different environmental conditions is crucial. Vessel operations are also a key consideration when designing a successful coating. The operational profiles of different ships can be quite variable, ranging from constant use at high speeds to frequent and extended periods of docking (6). A truly universal coating, therefore, is required to perform well in both static and dynamic conditions. Lastly, these coatings are expected to perform on these ships for years if not decades, so they need to be long lasting and quite durable without the need for constant reapplication. All these factors need to be taken into consideration in the design of a successful material.

Today, coatings are generally designed to act in two ways: as antifouling (AF) and fouling release (FR) materials, as demonstrated in **Figure 1**. An AF coating prevents the accumulation of biomaterials on a surface. A perfect AF coating should completely prevent the growth of marine species on submerged surfaces. However, it is inevitable that some species will be able to settle on these surfaces, often promoting the settlement of subsequent organisms (25–27). In this case, a FR surface is desirable because they are designed to weaken the adhesion of fouling organisms to a surface, facilitating easy removal. As we explore in the following sections, these solutions to the fouling problem involve two different mechanisms that can be at odds with one another, requiring materials with wildly different surface properties. Researchers today are exploring novel ways to incorporate both mechanisms into their systems.

In this review, we explore the key characteristics required in the design of successful nontoxic polymer-based AF and FR coatings. As mentioned above, with the multitude of requirements placed on these coatings, it is in the research field's best interest to explore multicomponent coatings that can act through a variety of mechanisms and have broad spectrum effects. We explore what the field has identified as the most vital components for consideration in the design of these coatings, where these components intersect, and where there are still gaps that will direct the research field in the future.

Antifouling: a surface designed to prevent the attachment of organic matter and aquatic organisms to a surface

Fouling release: a surface designed to facilitate easy removal of organic matter and aquatic organisms from a surface

FACTORS INFLUENCING ANTIFOULING AND FOULING RELEASE BEHAVIOR

Surface Chemistry

Amphiphilic:

a compound or surface that possesses both hydrophilic and hydrophobic components

Zwitterion: a neutral molecule that bears both a positive and negative charge

Probably the most important criterion in the design of AF and FR coatings is surface chemistry. Hydrophilic coatings, with high surface energies similar to water (72 mN/m), perform well as AF materials, as the interfacial energy between the surface and water is minimized. This removes any thermodynamic advantage from the adsorption of biomolecules, because it is energetically favorable for the surface to remain in contact with water rather than an amphiphilic biomolecule like a protein. Often these surfaces possess the ability to order water molecules, further stabilizing the hydration and enhancing the resistance to water displacement. In contrast, very hydrophobic, low-surface energy materials work well in FR coatings. These surfaces are designed to minimize interactions with biomolecules by eliminating the ability for strong polar interactions, like hydrogen or ionic bonding. Through dispersive interactions alone, biomolecules can adhere only very weakly to these surfaces, making their removal easier.

One of the most ubiquitous AF materials is poly(ethylene glycol) (PEG), which has found widespread use in a variety of different structures (28–31). Despite its excellent performance, PEG lacks long-term stability and is susceptible to oxidative and microbial degradation (32–35), so recent research has focused on identifying alternatives. Charged structures, such as zwitterions and polyelectrolytes, are one such alternative because they can form tightly bound and highly structured water layers (36). Current research trends include the incorporation of these charged components into complex structures and the study of the effects of charge on the behavior of different marine organisms.

Although hydrophilic surfaces have shown immense promise, they generally perform well only as AF coatings and do not readily release fouling organisms once adhered. Some marine organisms even possess the ability to displace water from a surface, rendering the hydration layer useless (37). In these instances, a FR coating is necessary. The relationship between surface energy and FR is well established and described by the Baier curve (**Figure 2**) (38–40). Plotting the relative adhesion of foulants versus critical surface tension, Baier found that there was a minimum between 20 and 25 mN/m that gave optimal FR. This range corresponds most directly to polydimethylsiloxane (PDMS) and fluorinated materials, which have been explored extensively for FR coatings (40, 41). Unfortunately, these coatings are highly hydrophobic and generally very poor AF materials because the high interfacial energy with water favors the adsorption of biomolecules. This limits their use in static applications, where there are no hydrodynamic shear forces to remove settled organisms.

Amphiphilic coatings are designed to combine the benefits of hydrophilic and hydrophobic surfaces. By imparting characteristics of both, researchers hope to create coatings that can act effectively as both AF and FR materials. Micro- or nano-domains of hydrophilic and hydrophobic components create an ambiguous surface for marine foulers, so they cannot sense a large enough hospitable surface to attach or adhere to (42–44). A variety of tactics are used to achieve amphiphilic surfaces, which Galli & Martinelli (45) recently reviewed comprehensively. Amphiphilic surfaces offer a convenient route to achieving AF/FR coatings and currently constitute a large portion of current research.

Physical Properties

Although surface energy plays an important role in AF and FR behavior, the physical properties of a surface are also significant. Barnacle removal can be modeled by the adhesion of a rigid solid

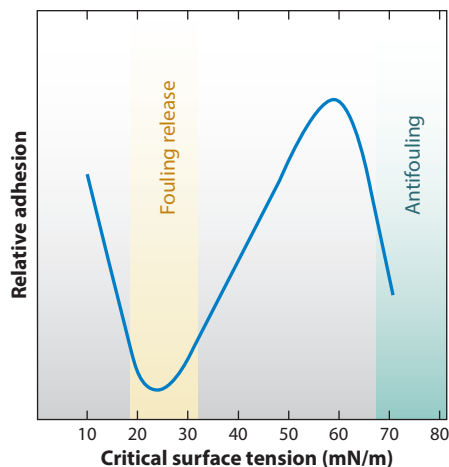


Figure 2

A Baier curve plotting the degree of fouling versus critical surface tension. There is a minimum between 20 and 25 mN/m where optimal fouling release is achieved and a trend toward a minimum above 70 mN/m for highly hydrated materials, which generally perform well as antifouling materials. Adapted with permission from Reference 40.

attached to an elastomeric film, where the critical pull-off force is proportional to $(E\gamma)^{1/2}$, where γ is the surface energy of the material and E is the modulus. As the modulus of the material is lowered, the critical pull-off force is minimized (46). PDMS, with both low modulus and low surface energy, is ideal for FR applications. Low-modulus materials have also been shown to induce abnormal growth in barnacle basal plates, visible in both their shape and size (47, 48). Although this relationship does not hold directly for soft foulers, there is still a strong correlation between low modulus and improved FR (49).

Film thickness has also been identified as an important factor in FR materials, and there exists an inverse relationship between the thickness and the critical pull-off force: As film thickness increases, adhesion strength decreases (46). This applies to both hard and soft foulers, with lower critical removal force of a barnacle from a 2-mm coating than either a 0.1- or 0.5-mm coating and more than doubled release of *U. linza* sporelings when coating thickness was increased from 16 to 100 μm (49, 50).

Another important factor is topography, which plays a significant role in how marine foulers interact with, settle on, and adhere to a surface. Effective coatings have been developed based on these behaviors, but it requires balance of size, geometry, and spatial arrangement of features to ensure the widest applicability (51). Much of the research in this area takes design cues from natural AF systems, such as Sharklet AFTM, based on the dermal denticles of shark skin. Composed of diamond-shaped scales with a ribbed pattern made from a PDMS elastomer (PDMSe), Sharklet AFTM reduces *U. linza* settlement up to 86% in laboratory assays (52). Unfortunately, field studies have shown that topography alone cannot combat fouling in complex marine ecosystems (53, 54), but surface features are still important in AF/FR coatings. Surface roughness is known to increase settlement and adhesion strength of biomolecules, including the proteinaceous adhesives used by many marine organisms (55). Highly textured or very rough materials present the opportunity for fouling organisms to settle within and between topographic features, thereby protecting them from the hydrodynamic forces (56, 57). Not only that, but roughened surfaces provide greater surface area for adhesive cements to adhere and mechanically interlock with (46). Despite the fact

that topography cannot act as the sole mechanism, it is still a key characteristic that must be taken into account when designing materials.

Durability and Attachment

An aircraft carrier can have a hull area of up to 160,000 ft² (58), so even if a coating satisfies all the above criteria, it will not be practical if it has a short lifetime. The stability must be considered in the design of any new chemistries or surface structures for AF/FR applications. A large sector of research focuses on increasing durability, from using networks to self-healing materials. Although not included in the scope of this review, the use of nanocomposites for increasing the durability of polymeric coatings is a common and highly successful practice, and Selim et al. (59) provided a thorough review of this topic recently. New attachment chemistries are also being explored. Phenolic chemistries offer excellent and nonspecific affinity for most surfaces, facilitating effective attachment to almost any type of substrate. One of the most ubiquitous, 3,4-dihydroxy-L-phenylalanine (DOPA), is inspired by a bioadhesive used in the feet of marine mussels. Its allure for marine AF applications derives from its ability to attach and adhere to wet surfaces in environments dominated by high salinity and turbulent waves, promising an effective coating strategy for use on marine hulls. More recently, other polyphenolic compounds, such as tannic acid, are garnering research interest. These compounds can strongly bind to a surface, while also possessing the ability to participate in hydrogen bonding, hydrophobic interactions, and cation binding (60). Ultimately, to ensure that the coating is appropriate for marine applications, researchers need to optimize durability and attachment schemes.

RECENT DEVELOPMENTS IN ANTIFOULING AND FOULING RELEASE COATINGS FOR MARINE APPLICATIONS

Today coatings are being designed with surface chemistry, mechanical properties, durability, and substrate attachment in mind. In the design of broadly applicable coatings, researchers are using different chemistries, synthesis methods, and structures to account for all of the above criteria, while further exploring the fundamentals of why these aspects provide successful materials. We discuss the different routes researchers are taking to address some or all these attributes, as summarized in **Figure 3**, and the extent of success that the subsequent coatings have achieved. For reference, see the sidebar titled Coating Evaluation Strategy and Glossary of Laboratory Assay Fouling Organisms for a brief discussion on the process of testing a coating and an overview of some of the most commonly used species for laboratory assays.

Brushes

Hydrophilic polymer brushes composed of tightly packed chains covalently anchored to a surface have shown excellent AF properties owing to their ability to form a tightly bound water layer and high resistance to compression from steric repulsion of neighboring chains (61). Investigation of how different brush characteristics affect interactions with fouling organisms is garnering increasing interest. Yandi et al. (55) investigated hydration and chain entanglement of copolymer brushes of 2-hydroxyethyl methacrylate and PEG methacrylate ranging from 50 to 1,000 Å thick and found that brush thicknesses of 200 and 400 Å showed the best AF performance. This behavior was attributed to the optimal hydration capacity of these brushes; thinner films cannot hold enough water, whereas the hydration of thicker films is reduced because of crowding and chain entanglement, so water is not able to penetrate the brush as efficiently. Conversely, when

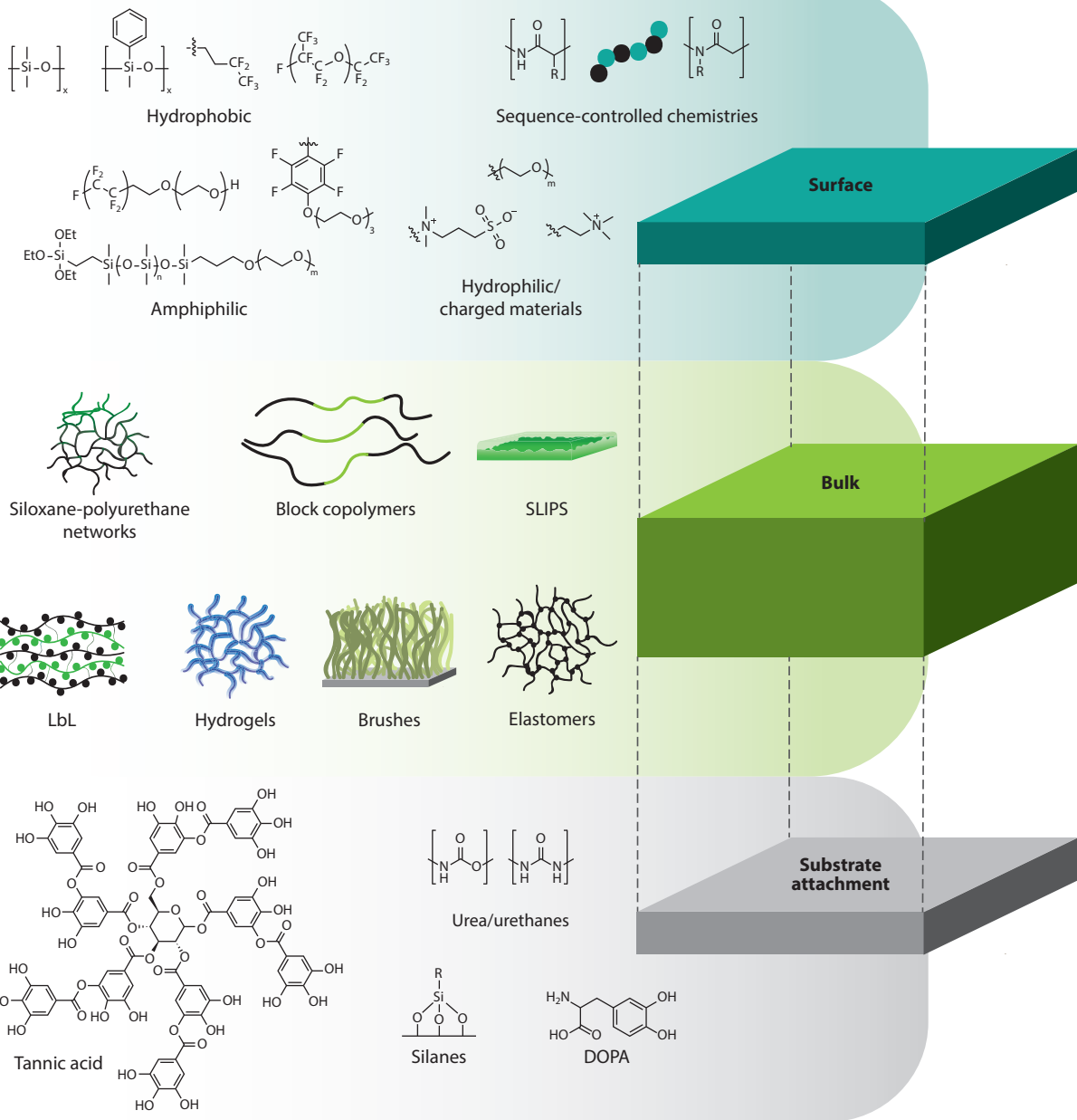


Figure 3

Overview of current research trends in the development of polymer-based environmentally friendly antifouling and fouling release coatings. The diagram shows the most popular surface chemistries, bulk structures, and compounds and structures used to control substrate attachment. Abbreviations: DOPA, 3,4-dihydroxy-L-phenylalanine; LbL, layer-by-layer; SLIPS, slippery lubricant-infused porous surfaces.

COATING EVALUATION STRATEGY AND GLOSSARY OF LABORATORY ASSAY FOULING ORGANISMS

In the development of new antifouling and fouling release technologies, coating materials usually will be tested first in laboratory assays, then in short-term field tests, and lastly in long-term performance assays that can be multiyear studies (126). Rapid laboratory assessment of coating activity can help to quickly identify potentially viable materials. Single-organism laboratory assays also allow for precise control of conditions, which can help to better elucidate the exact mechanism of action. Successful coatings can then be adapted and subsequently tested in short- and long-term field tests.

Listed below are some commonly used organisms for laboratory assays, categorized by their fouling type (126, 127). Although by no means comprehensive, this will hopefully serve as a reference to the reader for the remainder of this review.

Soft microfoulers

Marine bacteria: *Cellulophaga lytica*, *Halmononas pacifica*, *Pseudomonas*

Diatoms: *Navicula incerta*, *Amphora coffeaeformis*, *Phaeodactylum tricornutum*, *Cylindrotheca closterium*

Microalgae: *Chlorella vulgaris*

Soft macrofoulers

Algae: *Ulva linza*

Hard macrofoulers

Mussels: *Perna viridis*, *Geukensia demissa*

Barnacles: *Balanus amphitrite*, *Balanus improvisus*, *Semibalanus balanoides*

Bryozoans: *Bugula neritina*

Tube worms: *Hydroides elegans*

Kuliasha et al. (62) grew copolymers of acrylamide, acrylic acid, and methyl acrylate of 2–8-nm thicknesses, with graft molecular weights in the range of 100–1,400 kg/mol, little difference was found in the settlement or removal of *U. linza* and *Navicula incerta* between coatings. Likely the reduced thickness, and amphiphilic nature, of these materials accounts for the lack of observed differences. Quintana et al. (63) investigated the effects of changing the polymerizable group in a sulfobetaine polymer brush and found little difference in the reduction of barnacle cyprid and *Amphora coffeaeformis* settlement. The comparable performance indicates that the brush backbone does not play a significant role in the performance of these materials, and the driving force is the zwitterionic moiety.

Because of stress induced by the stretched and crowded nature of brushes, the ultimate stability of these materials is of great concern (64, 65). Even when grown using a hydrolytically stable backbone, surface attachment points represent a weakness in the brush. Typically, brushes are grown from a surface using a silane initiator whose siloxane bond presents a hydrolytic instability (64). Quintana et al. (66) grew poly(sulfobetaine methacrylamide) (PSBMAM) polymer brushes with a hydrophobic lower block designed to exclude water from the tethering point of the brush and protect it from hydrolysis. A polystyrene block prevented degrafting even after 12 weeks in artificial seawater (ASW), whereas the control PSBMAM brush completely degrafted after 4 weeks. Phenol-based initiator systems have also been used in place of silane initiators, providing more stable attachment while also increasing the ease of application of coatings. Pranantyo et al. (67) used a tannic acid initiator to grow several types of charged brushes from stainless-steel surfaces, which were able to resist settlement of marine bacteria, diatoms, and barnacle cyprids. Brushes were able

to maintain stability under constant flow of seawater for 14 days with no detectable damage to the surface coating. Xu et al. (68) developed a system to post-functionalize polydopamine coated on stainless steel with thiol and azide functionalities to assemble binary polymer brushes of poly(2-methacryloyloxyethyl phosphorylcholine) (PMPC) and poly[2-(methacryloyloxy) ethyl trimethylammonium chloride] (PMETAC) via orthogonal click chemistries. PMPC brushes were grown using a thiol-ene photopolymerization, and PMETAC polymer chains were grafted to the surface by an azide-alkyne click reaction. This surface successfully inhibited settlement and adhesion of two types of bacteria and *A. coffeaeformis*, even after aging for 30 days in seawater.

As the application of materials bearing charge is becoming more widespread, the influence of surface charge on fouling organisms is garnering significant research interest. Yandi et al. (69) used four brush types, cationic, anionic, neutral, and zwitterionic, to investigate the influence of surface charge on several fouling organisms. In laboratory assays, the anionic, neutral, and uncharged brushes outperformed cationic brushes, with neutral and uncharged brushes showing the best broad-spectrum AF properties. Higaki et al. (70) saw similar behavior when examining the settlement behavior of barnacle cyprids, mussel larvae, and marine bacteria on zwitterionic, cationic, and hydrophobic brushes, with the zwitterionic brushes showing the best AF performance. Although the cationic brushes were able to inhibit barnacle cyprid settlement, they performed equal to or worse than hydrophobic brushes in mussel and marine bacteria assays. Several marine fouling organisms possess negatively charged cell walls, so positively charged surfaces can promote their settlement. Despite this, positively charged quaternary ammoniums are still popular components because of their antimicrobial activity. In a study investigating the algicidal nature of cationic poly[2-(dimethylamine)ethyl methacrylate] (PDMAEMA) brushes, Yandi et al. (71) found that despite performing relatively poorly in settlement assays against *U. linza* spores, PDMAEMA brushes had strong antialgal properties, causing pseudosettlement and ruptured spore bodies in 40% of settled spores, and significantly inhibited growth into sporelings.

Polymer brushes are also being used to modify the surface chemistry of traditional FR materials. Zhang et al. (72) grew brushes on wrinkled PDMS_e surfaces with feature sizes between 1 and 3 microns. Samples tested in a turbulent flow chamber against the diatom *Chlorella vulgaris* showed that all functionalized PDMS_e surfaces were able to reduce settlement and adhesion, with PEG and zwitterionic brushes able to reduce settlement to less than 1% of the PDMS control. In laboratory assays, the wrinkled surfaces outperformed their flat counterparts, but field immersion tests showed little difference between the two. Kuliasha et al. (62) saw similar results using a copolymer of acrylamide, acrylic acid, and methylacrylate grown on PDMS_e substrates that were either topographically smooth or patterned with Sharklet AFTM. Overall, the films performed well against *U. linza*, reducing settlement by 85–92% and improving *N. incerta* fouling removal by 55% compared with PDMS_e controls, but no statistical difference was seen between the two topographies. These results emphasize surface chemistry's larger role in material development than topography.

Sequence-Controlled Chemistries

As amphiphilic surfaces have gained attention for building robust AF/FR surfaces, there is growing interest in controlling precise architectures and compositions of hydrophobic and hydrophilic components. One tool that has emerged as a viable candidate is sequence-controlled chemistries. Calabrese et al. (73) synthesized nonnatural amino acids with either an alkyl or PEG side chain that were used to build amphiphilic oligopeptides. When attached to a polystyrene-*b*-poly(dimethylsiloxane-*co*-vinylmethylsiloxane) [PS-*b*-P(DMS/VMS)] block copolymer backbone, oligopeptides markedly lowered contact angles and improved AF/FR performance against *U. linza* compared with an unmodified control. In a separate study, the oligopeptide composition

was expanded to incorporate a fluorinated hydrophobic group (74). Overall, fluorinated oligopeptides were better at surface segregating than their alkyl counterparts, resulting in a greater population of oligopeptides at the surface. Modification with either oligopeptide reduced settlement of *U. linza* compared with the unmodified control but increased settlement of *N. incerta* and reduced removal against both species. No statistical difference was seen between the fluorinated and alkylated oligopeptides against *U. linza*, whereas the alkylated materials performed better in both assays of *N. incerta*. Although fluorine serves as a better chemistry to bring the oligopeptides to the surface, it needs to be used sparingly, especially in an already hydrophobic coating.

Although these coatings show promise for introducing amphiphilicity, peptides can be expensive and synthetically challenging to produce, especially when requiring the production of non-natural amino acids. Peptoids are a structural analog of peptides that use a submonomer synthetic route that allows for higher yields and simple incorporation of more diverse chemistries (75). In addition, the *N*-substituted glycine backbone is resistant to proteolytic degradation, making it better for use in biological applications (76). **Figure 4** provides comparison of the structure and synthesis route for peptoids and peptides. Van Zoelen et al. (77) systematically varied the length and sequence of amphiphilic oligopeptoids with a methoxy and fluorinated side group to explore sequence and spacing of hydrophobic and hydrophilic components in a polystyrene-*b*-poly(ethylene oxide-*co*-allyl glycidyl ether)-*b*-polystyrene [PS-(PEO/AGE)-PS] polymer system. In laboratory assays against *U. linza*, location of the fluorinated group was key to AF performance, with fluorinated groups placed closer to the backbone leading to the lowest settlement. The number of fluorinated units had more of an effect on FR, with a single fluorinated unit providing the highest removal. Sum frequency generation spectroscopy confirmed the importance of sequence and composition, as the number and position of fluorinated units heavily influenced surface composition of coatings underwater (78). The oligopeptoid sequence was also shown to play a role in the ordering of interfacial water, which affects the hydration layer at the polymer-water interface.

Most recently, Patterson et al. (79) directly compared the performance of peptides and peptoids, attaching each to a PS-P(DMS/VMS)-PS or PS-P(EO/AGE)-PS backbone. Overall, behavior was mostly governed by polymer backbone, which has been observed elsewhere (74), but performance in both settlement and removal assays was generally enhanced in peptoid-containing coatings. On the PDMS backbone, the oligopeptoid structure showed considerably lower *U. linza* adhesion strength with up to 95% removal, which the authors attribute to the peptoid backbone lacking a hydrogen bond donor. *N. incerta* settlement on the oligopeptoid-functionalized PEO was also noteworthy, with zero observed settled cells.

Modified Elastomeric Networks

Because PDMSes have shown promise as FR materials, owing to their low modulus and surface energy, methods to enhance their AF properties are currently being explored. One simple and synthetically practical route is the inclusion of additives in the elastomeric matrix. The addition of silicone oils, for example, has proven excellent for FR, enhancing slipperiness and reducing the adhesive strength of fouling organisms (80, 81). Unfortunately, the addition of silicone oil alone still produces highly hydrophobic coatings, which are not ideal at preventing biofilm buildup. Hydrophilic or amphiphilic additives can be used to modulate the surface chemistry and improve AF performance. **Figure 5** provides an overview of the discussed additives.

Martinelli et al. (82) used amphiphilic side chains composed of a triethyleneglycol-modified pentafluorostyrene (EFS) and a PDMS-modified vinyl benzene that improved removal of *U. linza* by at least tenfold compared with PDMS controls even at wall shear stresses as low as 13 Pa. Additives with the highest EFS content improved AF capacity by more than 3 times compared

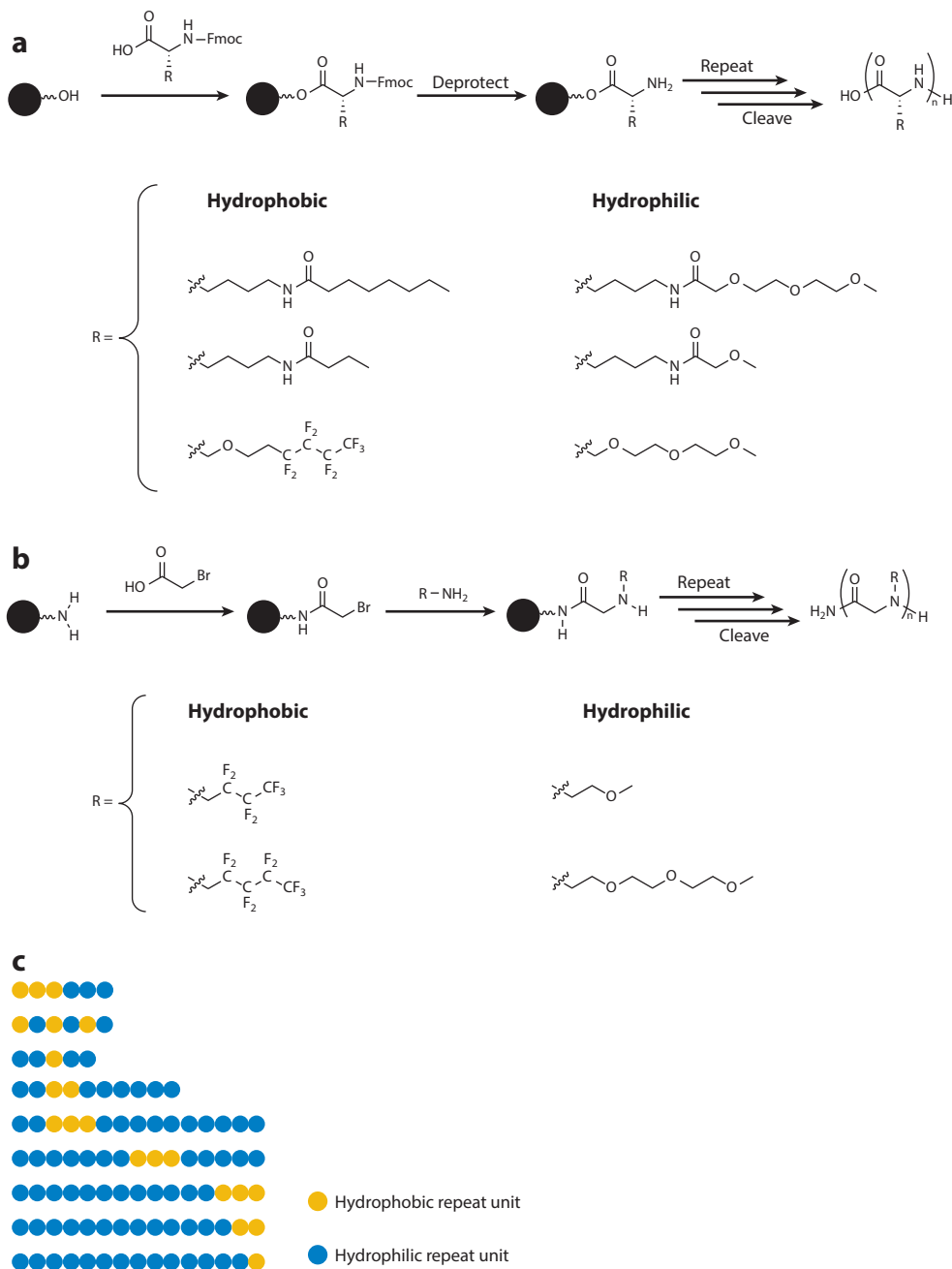


Figure 4

Solid phase syntheses for peptides and peptoids. (a) Peptide synthesis uses amino acids with a chiral center to which the desired side group is attached. The use of a protecting group is necessary to achieve attachment to the solid support only at the carboxylic acid, so after every attachment step a deprotection must be performed. (b) Peptoid synthesis employs a submonomer approach, so the carbonyl and side group are incorporated in two separate steps, which allows for a more diverse range of side groups and does not require a protecting group. The different hydrophilic and hydrophobic structures employed in the peptide and peptoid synthesis are also included. (c) Examples of sequences used, demonstrating the manner in which authors probe the sequence and compositional effects of different hydrophobic and hydrophilic moieties (73, 74, 77–79).

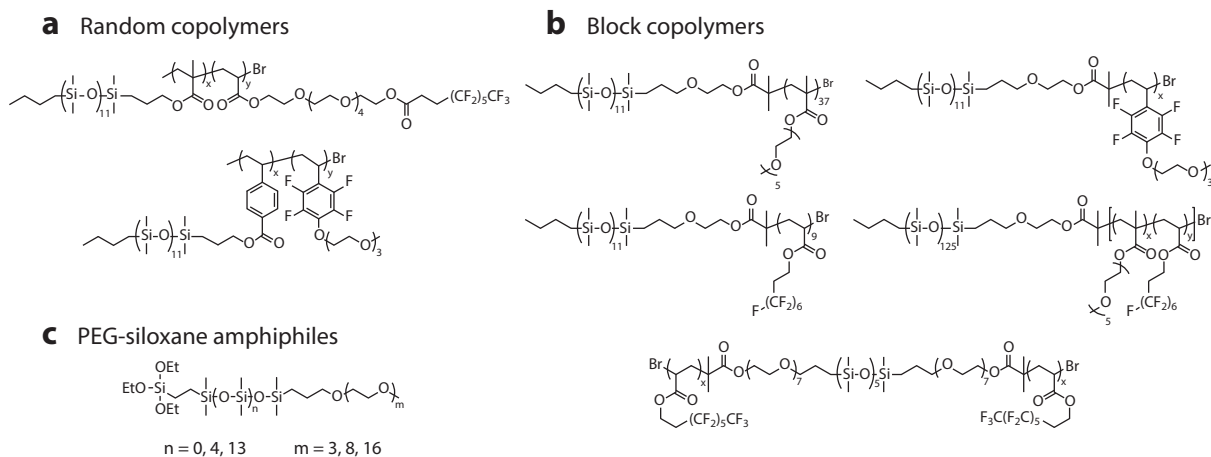


Figure 5

Structures of amphiphilic additives used in elastomeric matrices, including noncovalently included (a) random (82, 83) and (b) block copolymers (84–87), as well as (c) covalently linked poly(ethylene glycol) (PEG)-siloxane amphiphiles (88–92).

with PDMS controls. In another study, Galli et al. (83) used a random copolymer consisting of a methacrylate with mixed side chains of PEG and a fluoroalkyl in combination with a methacrylate possessing a siloxane side chain. Despite the PEG and fluoroalkyl being introduced in the same side chain, X-ray photoelectron spectroscopy (XPS) showed different ratios of CF_2/OCH_2 depending on the composition of the additive, with 25–35% more surface fluorine than theoretically calculated. Higher ratios of CF_2/OCH_2 at the surface resulted in higher settlement densities of *U. linza* spores and lower sporeling removal. Van Zoelen et al. (77) also observed this trend; in their study, fluorine assisted in bringing PEGylated components to the surface, but higher concentrations at the coating surface proved to be detrimental to overall performance.

The use of block copolymers facilitates more precise control of incorporation of different components and their placement within the polymer structure. For example, a terminal siloxane block can help to facilitate dispersion of additive in the bulk matrix, limiting the possibility of phase separation. Diblock copolymers of either a fluorinated or PEGylated acrylate grown from a PDMS macroinitiator showed vastly different performance in fouling assays (84). Whereas the PEGylated block copolymer improved removal of *U. linza* from 15% to 90%, the fluorinated materials performed no better than controls. In assays against barnacle cyprids, however, where the PEGylated material improved resistance to settlement by 50% compared with controls, the fluorinated coatings completely inhibited settlement. In a separate study, a PDMS macroinitiator was used to grow a random copolymer of fluorinated acrylate and PEGylated methacrylate. Improvements were achieved in the performance against *U. linza*, where close to 50/50 ratios of incorporation of PEGylated and fluorinated monomer facilitated removal close to 100% (85). These coatings were not tested against barnacle species, but in a separate study, in which block copolymers of PDMS and EFS were grown, the amphiphilic additives were able to improve AF characteristics compared with controls, especially at longer EFS chain lengths where settlement of barnacle cyprids of two separate species was completely inhibited (86).

Martinelli et al. (87) synthesized a pentablock copolymer consisting of PDMS, PEG, and a fluorinated acrylate. Against *U. linza* settlement, pentablock-containing coatings had higher settlement than PDMS controls, likely owing to high amounts of fluorine at the coating surface. In removal assays, pentablock materials gave significantly better removal of up to 70%, a 17-fold

increase over controls. These results further emphasize the importance of finding the correct balance of hydrophobic and hydrophilic components, especially when using fluorine, to achieve the most broad-spectrum AF/FR performance.

Low-molecular weight PEG-siloxane additives with crosslinkable triethoxy silane end groups, which covalently incorporate the amphiphilic additives to the PDMSe matrix, have also been explored. Systematic studies of PEG and siloxane length revealed how chain length controls performance. Hawkins et al. (88, 89) used PEG of constant length while modulating siloxane length. They found that increased siloxane tether length reduced biofilm formation by almost three orders of magnitude in laboratory assays and minimized diatom biofilm formation after six weeks in static immersion field tests. The authors attribute this to increased configurational mobility and solubility of the amphiphile in the matrix conferred by a longer siloxane segment. Varying the PEG length also impacted performance: An eight-unit PEG chain performed better than shorter or longer chain lengths, likely owing to a combination of sterics and solubility; shorter PEG segments cannot generate enough hydrophilicity at the surface, and longer segments have limited mobility in the matrix and are more prone to phase separation (90, 91). Also shown to play a key role was the amphiphile loading in the elastomeric coating, with higher loadings generally enhancing performance (90, 92).

Charged Networks

Hydrogels, hydrophilic network materials that have a high capacity for absorbing and holding water, have excellent promise as AF materials. Research into some marine organisms' response to hydrogel coatings has shown that some barnacle species will develop abnormal basal plates and have limited growth when settled on hydrogel materials because of their low modulus (47, 93). However, their poor substrate adhesion and low durability limit their application in marine environments, so recent efforts have focused on increasing stability. Jiang et al. (94) formed tough double network hydrogels from charged poly(trimethylammonium methacrylate) or poly(3-sulfopropyl methacrylate) strengthened with a second network of polyacrylamide. Formation of double networks improved modulus up to 5 times and maximum compressive stress up to 11 times. The negatively charged and neutral double networks were able to maintain less than 10% biofilm coverage, whereas the positively charged networks showed much higher settlement, in agreement with earlier studies done on charged materials. Yang et al. (95) used a similar strategy generating several semi-interpenetrating networks of polyacrylamide incorporating different amounts of charged, charge-neutral, and uncharged monomers interlaced with a polyvinyl alcohol network. Incorporation of polyvinyl alcohol was able to increase the elastic modulus by close to an order of magnitude. The best AF performance was achieved with the zwitterionic and PEG-containing monomers, with reduction in algal settlement of up to 20 times. Increased loading of the hydrophilic monomers did result in decreases in the material's elastic modulus, indicating the need for fine tuning of surface chemistry alongside mechanical properties.

Charged layer-by-layer (LbL) coatings are easy, fast, and cheap to assemble, requiring only alternating application of oppositely charged polyelectrolytes, usually by dip or spray coating. Zhu et al. (96) used a crosslinked LbL system made up of a polyanionic carboxylate polymer assembled by dip coating onto a silicon wafer in an alternating fashion with polycationic branched polyethyleneimine. Covalent crosslinking between the carboxylate and amine side groups stabilized films to prevent dissolution. These films reduced barnacle cyprid settlement by more than 85% compared with a silicon control. The application of the LbL system reduced surface roughness compared with the substrate from 1.8 to 0.65 nm. When an amphiphilic polyanionic material was made using a commercial fluorinated surfactant, Zonyl FSN-100, the films tested

against marine bacterium *Pseudomonas* were successful at almost completely inhibiting bacterial adhesion (97).

One drawback to polyelectrolyte LbL coatings is the excess surface charge that is governed by the last deposited material. The effects of varying surface charge were explored by tuning pH during the LbL assembly process (98). As already discussed, positively charged surfaces tend to be more attractive to marine species, which is true for *Pseudomonas*, for which positively charged surfaces promoted bacterial fouling, whereas negatively charged surfaces were better able to resist it. Conversely, for barnacle cyprids, positively charged and neutral surfaces were best able to resist settlement, whereas negatively charged surfaces promoted it.

Zwitterions offer a promising alternative to PEG, but there are drawbacks that make them difficult to implement: Monomers can be expensive, they are notoriously difficult to work with in organic solvents, and they can even interfere with or be unstable during polymerizations (99). Because of the high cost of zwitterionic monomers, Ventura et al. (100) explored incorporation of only small quantities of zwitterions (0–20 mol%) to form an amphiphilic coating. Variable amounts of carboxybetaine methacrylate (CBMA) or sulfobetaine methacrylate (SBMA) were added into a lauryl methacrylate network. Whereas the incorporation of CBMA improved removal of biofilm formers, SBMA decreased AF/FR performance, especially at higher loadings. Although the SBMA coatings exhibited higher surface roughness, they also had significantly lower modulus values (<20 MPa) than CBMA coatings (>80 MPa), which would usually favor FR. These results are completely opposite to trends that were previously seen, where pure SBMA self-assembled monolayers tend to perform better than pure CBMA (101). This suggests that functional group concentration is important for zwitterions, while also highlighting the interplay of surface roughness and modulus and the necessity to balance these attributes.

Recently researchers have explored novel ways to generate zwitterions in network materials to circumvent their synthetic challenges. Wang et al. (102) designed an amphiphilic network material containing PDMAEMA that was functionalized post-application to form carboxybetaine zwitterions by quaternizing with bromopropionic acid. Zwitteration lowered the settlement of the diatom *Phaeodactylum tricornutum* by more than 50%. Xie et al. (103) took this a step further, creating a degradable network material that generated zwitterions in situ using a polyester and a tertiary carboxybetaine ester. When in contact with seawater, the tertiary carboxybetaine ester protonates at the tertiary amine and hydrolyzes the ester to form a zwitterion, while the polyester in the backbone hydrolyzes to erode the surface and facilitate continuous renewal of zwitterions. This material exhibits excellent bacterial resistance, indicating the simultaneous generation of zwitterions and AF efficacy in situ, while also representing a simple synthetic methodology for the incorporation of zwitterions in a network material.

Polyurethane and Polyurea Networks

Although PDMS_e shows great promise as a FR material, it lacks mechanical durability and does not adhere well to substrates, putting it at risk of delamination. The Webster research group has developed a system to solve these issues using a siloxane-polyurethane (SiPu) network that self-stratifies, providing a PDMS-enriched surface and a polyurethane bulk. This improves durability and increases adhesion to substrates but results in very hydrophobic coatings (104, 105). Researchers are now introducing hydrophilic components to improve AF properties and tune FR for a wider range of organisms, which are summarized in **Figure 6**. Bodkhe and colleagues (106, 107) first did this by introducing carboxylic acids and then PEG onto the PDMS backbone, both of which showed promise for developing amphiphilic coatings. Building upon this, Galhenage et al. (108) constructed coatings with varying loadings and molecular weights of PDMS and PEG

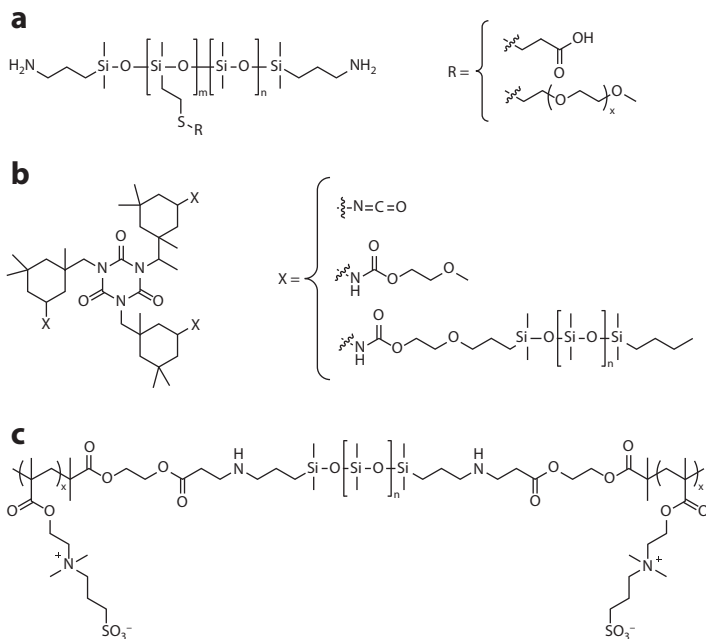


Figure 6

Methods for incorporation of hydrophilic groups in siloxane-polyurethane (SiPu) networks, including (a) attachment of hydrophilic side groups to a polydimethylsiloxane (PDMS) backbone; (b) creation of an isocyanate prepolymer with PDMS, poly(ethylene glycol), and still-functionalizable isocyanate moieties; and (c) creation of a triblock copolymer with a PDMS central block and hydrophilic zwitterionic terminal blocks (106–109).

to tune the balance of hydrophobic and hydrophilic components. These coatings showed broad-spectrum efficacy that was comparable to or better than commercially available coatings, with greater than 90% of biofilm removal and very low adhesion strength for barnacles and mussels. The PDMS chain length played a large role in hard fouler behavior, whereas soft fouler behavior depended more on PEG chain length; in both cases, longer chains provided better removal.

Zwitterions have also been used as the hydrophilic component in SiPu coatings. Bodke et al. (109) used PSBMA as part of a block copolymer with PDMS that was incorporated into the polyurethane network. Although these coatings had excellent FR properties against the marine bacterium *Halomonas pacifica* and *N. incerta*, the performance was less pronounced against *Cellulophaga lytica* and *U. linza*. The authors attribute this mixed performance to improper balance of hydrophobic and hydrophilic components. As Ventura et al. (100) demonstrated, the incorporation of low loadings of SBMA into a hydrophobic methacrylate-based network material did not seem to impart the same level of AF capacity as pure zwitterionic materials, stressing the importance of concentration when considering the use of zwitterions in amphiphilic materials.

Polyureas have also been used to introduce durability and increase substrate adhesion in network coatings. Liu et al. (110) designed a hydrophobic polyurea-based FR coating using PDMS, an alkyldiamine, and a diisocyanate. The storage modulus of the materials increased with incorporation of the polyurea from 1.5 MPa to greater than 80 MPa. In dynamic tests simulating a ship cruising speed of 18 knots, the polyurea coatings showed excellent substrate adhesion, whereas PDMSe coatings delaminated. Polyurea coatings provided the highest fouling removal of *N. incerta* in laboratory assays, although AF performance was on par with controls. Elsewhere, the

polyurea functionality was used for a PDMS coating with self-repairing capabilities. A diamine-terminated PDMS was reacted with a diisocyanate to form a PDMS-based polyurea (111). This material possessed the ability to reversibly crosslink at room temperature, owing to the hydrogen bonding between urea functionalities, and demonstrated excellent substrate adhesion, with stratification of PDMS at the surface and urea at the substrate interface similar to SiPu coatings. This material possessed the ability to self-heal even in ASW, recovering nearly full ultimate strength and elongation 48 h after damage.

SiPu coatings can also be modified with silicone oils to further enhance FR capacity. Galhenage et al. (112) tested the effects of inclusion of silicone oil in a PDMS–polyurethane system by synthesizing coatings in the presence of a variety of phenylmethyl silicone oils. XPS analysis showed an enriched silicone oil layer at the surface of up to 20 nm. In laboratory assays, the coatings significantly reduced adhesion strength for barnacles and mussels and improved FR properties against *U. linza*, but performance against biofilm formers, *N. incerta* and *C. lytica*, was no better than controls.

Slippery Lubricant-Infused Porous Surfaces

Another promising research area is slippery lubricant-infused porous surfaces (SLIPS), which are porous or roughened substrates coated in a lubricant that form a stabilized liquid interface. Lubricant is retained on the surface through capillary forces, sometimes enhanced by surface functionalization to increase surface affinity with the lubricant. SLIPS have shown immense power as AF/FR materials because they are a mobile interface, making it difficult for a fouling organism to sense and attach to them. Epstein et al. (113) demonstrated their capabilities using a system of porous polytetrafluoroethylene (PTFE) with fluorinated KrytoxTM lubricants, which exhibited a one-to-two order-of-magnitude improvement at preventing biofilm attachment compared with a PEGylated surface. Xiao et al. (114) expanded this system by infusing a methacrylate-based porous substrate with different fluorinated lubricants, which maintained a stable lubricant layer even after one month of continuous agitation in ASW. Coatings performed very well in preventing settlement of *U. linza* and barnacle cyprids, with performance on par with a PEGylated surface. Experimental coatings performed much worse than PDMS and PTFE controls in *U. linza* removal assays because sporeling roots were able to penetrate through the lubricant and attach firmly to the porous substrate.

Another concern for SLIPS is the loss of lubricant, which severely inhibits coating performance. Howell et al. (115) studied lubricant loss under flow and found that losses are on the order of only ng/cm² in a closed system and can withstand shear stresses up to 2.32 N/m². When an air–water interface is introduced, such as bubbles, the rate of lubricant loss increases by close to two orders of magnitude. A solution was developed in the form of a PDMS substrate possessing an underlying vasculature that could replenish the surface with lubricant (116). This system showed excellent performance, with only 10% biofilm coverage after 48 h, whereas the nonvascularized system had more than 90% coverage. The use of the PDMS-based substrates has also proven to be beneficial for developing robust FR surfaces. Amini et al. (117) used porous PDMS infused with a silicone oil that completely inhibited mussel settlement in laboratory assays and lowered their adhesive strength compared with commercial coatings by up to a factor of 4. When the coatings were tested in a 16-week field immersion test, the PDMS SLIPS showed the best performance by far, also markedly reducing soft fouler settlement. Although only polymer-based substrates are discussed here, SLIPS can be constructed on a variety of porous or roughened substrates. They have proven successful on metal-based surfaces for anticorrosion coatings and glass surfaces that are optically clear, demonstrating the broad applicability of this system (118–121).

CONCLUSIONS AND FUTURE OUTLOOK

Recent research identifies the biggest challenges the field is currently facing and indicates the directions in which future research will trend. First is the greater incorporation of charged materials into coatings. Although surface charge can improve effectiveness, especially in a laboratory setting, and can even provide antimicrobial benefits in the form of quaternary ammonium groups, the variability of species interactions with surface charge indicates that a neutral surface will provide the best broad-spectrum resistance. Zwitterions are a promising choice because of the stability and hydration they offer, but their effectiveness may be concentration dependent. Amphiphilic surfaces have proven to be one of the best routes to a dual AF/FR coating and are likely to continue to play a large role. At present, an optimal balance of hydrophobic and hydrophilic components appears to be the field's biggest challenge. Although a set of universal design rules would be ideal, there will be strong dependence on coating architecture and the type of hydrophobic and hydrophilic components used, and researchers need to thoroughly examine the interplay of these characteristics regarding their specific systems. PDMS-containing materials are still a large player in the research field, but research into how to best optimize their properties for broader applicability is still widespread, whether that comes in the form of amphiphilic additives or use in a SLIPS system. Lastly, the use of nonspecific initiators and attachment schemes like DOPA or tannic acid, SiPu networks, and self-healing materials offer a few routes to better improve the substrate adhesion and durability of these systems.

Another promising area of research is the inclusion of active chemistries to combat fouling. Several natural products have been identified that can inhibit quorum sensing, signaling that allows bacteria to control population density (122). Iron deprivation, using metal chelators like tannic acid, has proven effective in inhibiting bacterial growth and marine fouling (123, 124). Antioxidant additives have also been shown to reduce mussel adhesion strength by inhibiting radical pathways used in the formation of DOPA in their cements (125). Pairing of these active chemistries with previously discussed coating structures could help to actively deter and minimize adhesion of foulers by multiple mechanisms. A greater understanding of the marine organisms involved in biofouling, and the mechanisms they employ in settlement and adhesion, will be necessary for the development of truly effective strategies.

Ultimately, the real challenge facing the research community today is how to marry all of these attributes into one universal coating. Robust and durable coatings are needed, but they require low elastic modulus for optimal FR properties. Coatings must at once repel foulers from settling and minimize their adhesion, allowing them to be easily removed, which often necessitates a balance of hydrophilic and hydrophobic components. And these chemically, mechanically, and topographically complex materials need to be relatively easy to apply to a boat, ideally being suitably adapted into a paintable material. Even the most promising technologies addressed in this review still face considerable challenges before their ultimate implementation.

SUMMARY POINTS

1. In the design of coatings for antifouling and fouling release, surface chemistry, physical properties, durability, and substrate adhesion have all been identified as important considerations for building successful coatings. Coatings for marine applications present a unique challenge for researchers because of the harsh marine environment in which these materials are required to perform.

2. Antifouling surfaces are generally composed of hydrophilic materials. In the past, poly(ethylene glycol) was ubiquitous for the generation of a hydrophilic surface, but recent years have seen a surge in the application of charged materials, especially charge-neutral zwitterions, because they are more resistant to oxidative damage.
3. Fouling release surfaces are generally composed of low surface energy, hydrophobic materials to limit the polar interactions between foulers and the surface, making removal easier. Polydimethylsiloxane is one of the most successful fouling release materials because of its low surface energy and low elastic modulus, both of which promote fouling removal.
4. Amphiphilic materials have gained popularity as a method to incorporate the benefits of antifouling and fouling release mechanisms into one coating. Surfaces that possess both hydrophilic and hydrophobic domains present an ambiguous surface for foulers, so they are less likely to settle but also cannot adhere very strongly.
5. Physical properties of coatings are key in the design of a fouling release material. Elastic modulus and thickness play a key role in the performance of these coatings. Lower elastic modulus and thicker coatings promote fouling removal but must be designed precisely so as not to compromise durability and toughness.
6. Coating durability and strong substrate adhesion are important in developing coatings with long lifetimes. Phenolic chemistries, such as 3,4-dihydroxy-L-phenylalanine and tannic acid, increase coating adhesion to substrates, while also being nearly universally applicable, working to adhere to almost any surface. Polyurethane and polyurea coatings also offer an excellent alternative to generate better substrate adhesion while also creating more durable bulk material.

DISCLOSURE STATEMENT

The authors are not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

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